

Zero- Flow Engine

Beginner Setup Guide — Laptop / CPU Mode (no NVIDIA GPU)

Windows 11 · CPU mode, no graphics card needed · github.com/zerocold7/local-voice-flow

NOTE Have an NVIDIA, AMD or Intel graphics card? Follow the Install Guide instead — it covers every kind of PC, and the Hardware Guide explains what to expect from yours.

This guide takes you from a fresh Windows 11 laptop to a working Zero- Flow setup, **one step at a time** — written for someone who has never installed something like this before. It uses **CPU mode**, so **you do not need an NVIDIA graphics card**.

Zero- Flow does two things, entirely on your own machine:

- **Dictation** — press a key, speak English or Arabic, press it again, and your words are typed wherever your cursor is.
- **Reading aloud** — highlight any text, press **F4**, and a natural voice reads it to you.

Just follow the steps in order. Every command is copy-paste.

What you need before you start

- Your laptop, signed in to Windows 11.
- An internet connection — you'll download about **6–8 GB** total (one time only).
- About **30–45 minutes** (most of it is just waiting for downloads).
- Roughly **10 GB** of free disk space.
- Your **built-in microphone** and speakers (no extra hardware needed).

TIP No NVIDIA GPU? That's completely fine. This guide sets up **CPU mode**. It works exactly the same; your words just take about **1–4 seconds** to appear after you stop recording, instead of being instant.

Step 1 — Open PowerShell

PowerShell is the window where you'll type the setup commands.

- 1 Click the **Start** button (Windows logo, bottom-left).
- 2 Type the word **PowerShell**.
- 3 Click **Windows PowerShell** in the results. A dark window opens.

You'll type (or paste) one command at a time and press **Enter** after each. To paste a copied command, **right-click** inside the window.

Step 2 — Install the three required programs

You need three free programs: **Python** (runs the engine), **Git** (downloads the project), and **Ollama** (the local AI). Paste these three lines into PowerShell, pressing **Enter** after each:

```
winget install Python.Python.3.12
winget install Git.Git
winget install Ollama.Ollama
```

When they finish, **close PowerShell and open it again** (Step 1) so it notices the new programs. Then check they all installed:

```
python --version
git --version
ollama --version
```

Each line should print a version number (for example Python 3.12.x).

TIP "winget is not recognized"? Open the **Microsoft Store**, search for **App Installer**, and click **Update**. Then close and reopen PowerShell and try again.

TIP Typing python opens the Microsoft Store? Go to **Settings** → **Apps** → **Advanced app settings** → **App execution aliases** and turn **OFF** the two entries named "python". (Or simply type py instead of python everywhere in this guide.)

Step 3 — Turn on the microphone (do not skip!)

Windows hides the microphone from desktop apps by default. If you skip this, the engine will record **silence** and type nothing.

- 1 Open **Settings** (Start menu → the gear icon).
- 2 Go to **Privacy & security** → **Microphone**.
- 3 Turn **ON** "Microphone access".
- 4 Turn **ON** "Let desktop apps access your microphone".

Step 4 — Download the project

IMPORTANT Avoid OneDrive. New laptops often sync your **Desktop** and **Documents** folders to OneDrive, which can break file paths with confusing red errors during install. Put the project on the **C:\ drive** directly instead.

Back in PowerShell, run these lines. They put the project in a folder at the root of your C: drive (C:\local-voice-flow):

```
cd C:\
git clone https://github.com/zerocold7/local-voice-flow.git
cd local-voice-flow
```

Step 5 — Two small edits for a laptop without NVIDIA

Edit A — remove the GPU-only lines. Open the requirements file:

```
notepad requirements.txt
```

Find the **two lines that start with nvidia-** (nvidia-cublas-cu12... and nvidia-cudnn-cu12...) and delete just those two. Save with **Ctrl+S**, then close Notepad. They are only for NVIDIA cards and would waste ~1.2 GB.

IMPORTANT Delete **only** the two `nvidia-` lines. Keep the `kokoro` line near the bottom — that is the voice that reads text aloud.

Edit B — choose laptop-friendly settings. Create your settings file and open it:

```
Copy-Item .env.example .env
notepad .env
```

Find these four lines and change them so they read exactly like this, then save (Ctrl+S) and close:

```
WHISPER_MODEL_NAME="small"
WHISPER_DEVICE="cpu"
OLLAMA_MODEL_NAME="qwen2.5:3b"
READER_DEVICE="cpu"
```

TIP Why: the default speech model (`large-v3`) is built for a big graphics card — on a laptop processor it takes longer than you took to speak. `small` for speech and `qwen2.5:3b` for the AI fit comfortably in a 16 GB laptop, and the two `cpu` lines tell the engine not to look for a graphics card.

TIP Shortcut: instead of editing by hand, run `python check_hardware.py --apply`. It looks at your laptop and writes the right lines for you — and says so if yours has only 8 GB of memory and needs even lighter settings.

Step 6 — Install the engine's Python parts

These commands create a private folder named **venv** and download the needed libraries into it. Run them **in this order** (the last one takes several minutes — wait until the prompt returns):

```
python -m venv venv
.\venv\Scripts\python.exe -m pip install --upgrade pip
.\venv\Scripts\python.exe -m pip install -r requirements.txt
```

Step 7 — Download the local AI model

Ollama runs quietly in the background after install (look for its icon near the clock). Download a small bilingual model:

```
ollama pull qwen2.5:3b
```

This is about 2 GB, one time only. Lighter option: `gemma2:2b`. Higher quality but slower: `qwen2.5:7b`. Whichever you pull, put the **same name** in `OLLAMA_MODEL_NAME` in your `.env` (Step 5).

Step 8 — Start it for the first time

- 1 Open **File Explorer** and go to `C:\local-voice-flow`.
- 2 Double-click `Launch_Zero.bat`.

IMPORTANT If Windows shows a blue "Windows protected your PC" box, click **More info**, then **Run anyway**. This is just Windows being cautious about a new file — it's your own project file.

A console window opens. The **first** launch downloads the speech model (~500 MB) and the reading voice (~350 MB) — let it finish. Later launches load everything from your own disk and don't need the internet at all.

Dictation is ready when the window shows a box titled **Z E R O - F L O W E N G I N E** with the list of keys under it. **Reading aloud is ready** a little later, when this line appears:

```
[TTS] Voice model ready.
```

Leave that window open (you can minimize it). **Closing the window stops the engine.**

Step 9 — Test that it works

IMPORTANT Laptop Fn-key trap (important!). On most laptops, **F5–F10 are media keys** (volume/brightness) by default, so pressing **F5** alone won't reach the engine. Fix it once: press **Fn + Esc** to turn on **Fn-Lock** (this is the Lenovo/IdeaPad shortcut) — now the F-keys work with a single press. Or hold **Fn** together with the key every time (e.g. **Fn + F7**). If **Fn + Esc** doesn't work, look for a small padlock icon on the **Esc** key, or check your laptop's manual.

Dictation:

- 1 Open **Notepad** and click inside it so the cursor is blinking.
- 2 Press **F5** once (you'll hear a short beep), say "hello, this is a test", then press **F5** again.
- 3 After a second or two, your words appear in Notepad.
- 4 Try **F7** and speak Arabic the same way.

TIP Tap, don't hold. Each key starts recording on the first press and stops on the second. Holding it down makes it start and stop over and over.

Reading aloud:

- 1 Highlight a sentence in any window — the text you just dictated works.
- 2 Press **F4**. The voice reads it out. Press **Esc** to stop it early.

If nothing happens, see **Troubleshooting** below.

Step 10 — Hotkey reference

Press the key, speak, press it again. The key **forces** the language, so it's never misheard.

Key	What it does
F5	Dictate English — raw (exactly as spoken)
F6	Dictate English — polished (AI cleans grammar/fillers)
F7	Dictate Arabic — raw
F8	Dictate Arabic — polished
F9	Translate English → Arabic
F10	Translate Arabic → English
Shift + F1	Tidy up the learned-vocabulary file (the old list is kept as a backup)

Key	What it does
Shift + F2	Clear the debug log and dictation history
Shift + F3	Rewrite and fix the current line
F4	Read the highlighted text aloud (see Step 12)
Esc	Cancel the recording, or silence the reading

Spoken commands work inside any dictation:

- Start with **"new line"** to begin a new line, **"bullet"** for a • bullet point, or **"format code"** to wrap the text as code.
- End with **"and send"** to press **Enter** after it is typed (handy in chat apps).
- End with **"period"**, **"comma"** or **"question mark"** to type the real mark.
- In Arabic: every command has an Arabic equivalent — they are listed in the Arabic guide (SETUP-CPU-LAPTOP.ar.md) and in CONFIGURATION.md.

Step 11 — Everyday use: starting and stopping

- **Start:** double-click `Launch_Zero.bat` and minimize the window. (`Launch_Flow.bat` starts dictation only, without F4.)
- **Stop:** close that console window (click the **X**), or right-click the tray icon and choose **Exit Engine**.
- **Find the tray icon:** click the small ^ arrow next to the clock; the Zero- Flow icon lives there.
- **Ollama** starts automatically with Windows, so the AI features just work.

TIP Run it invisibly: double-click `Launch_Zero_Silent.vbs` to start with no window at all. To stop it, right-click the tray icon and choose **Exit Engine**. Stick with the visible .bat until you're comfortable — its window shows you what the engine is doing.

Step 12 — Reading text aloud (F4)

Highlight any text — a web page, a PDF, a chat message — and press F4. A local voice reads it out. Press Esc to stop it; to read something else, press Esc, highlight the new text and press F4.

F4 comes with `Launch_Zero.bat`, which this guide uses. (`Launch_Flow.bat` is dictation only.) The reader stays quiet while you are dictating, so it never talks over your microphone.

Right-click the tray icon to switch voice, to pause the reader (**Suspend Reader**), or to turn on **Smart LLM Cleaning**, which asks the local AI to strip page clutter (menus, cookie notices) before reading.

NOTE On a laptop with no NVIDIA card, expect a short pause before the voice starts. The voice model loads in the background when the engine starts; wait for Voice model ready. before your first F4. After that the voice starts once the first sentence is ready and keeps going smoothly while the rest is prepared. Reading a paragraph at a time rather than a whole page keeps that first pause short.

Troubleshooting

If this happens...	Do this
Nothing is typed when I speak	Check microphone permissions (Step 3). Press the key once to start and once more to stop — don't hold it. Look at the console window for a red error message.
The console says "Could not open the microphone"	The microphone is off, unplugged, or blocked — go through Step 3 again, then restart the engine.
python is not recognized	Close and reopen PowerShell. If it still fails, reinstall Python and tick "Add to PATH", or type py instead of python.
winget is not recognized	Update App Installer from the Microsoft Store, then reopen PowerShell.
It types in the wrong language	Use the matching key: F5/F6 for English, F7/F8 for Arabic. The key decides the language.
Hotkeys do nothing in one specific app	Right-click Launch_Zero.bat and choose Run as administrator .
Polish / Translate types my words unchanged	Make sure Ollama is running: type ollama list and check that the name in OLLAMA_MODEL_NAME in your .env is listed exactly (Step 7). A mismatch shows up in flow_debug.log as Ollama returned HTTP 404.
F4 does nothing	Make sure the text is highlighted first, and that the window has shown Voice model ready. The reader won't speak while a dictation is recording. Check you started Launch_Zero.bat, not Launch_Flow.bat.
Everything feels too slow	In .env set WHISPER_MODEL_NAME="base", save, and restart the engine.
SmartScreen blocks the launcher	Click More info , then Run anyway .
Pressing F5–F10 does nothing at all	Your laptop's F-keys are in media mode. Press Fn + Esc to enable Fn-Lock (Step 9), or hold Fn together with the key.
The laptop gets hot / fans get loud	Use the raw modes F5/F7 (they skip the AI). Confirm .env has small + qwen2.5:3b (not medium/7b). Drop to base for more cooling. See below.
Something else went wrong	Open flow_debug.log (dictation) or reader_debug.log (reading) in C:\local-voice-flow with Notepad — the last lines say what happened.

Keeping your laptop cool & fast

On a CPU laptop, **bigger models = more heat**. The biggest heat source is the AI model (Ollama), which works your CPU hard for a few seconds whenever you **polish** or **translate**. To stay cool:

- **For the least heat, use the raw modes F5 (English) and F7 (Arabic).** They type exactly what you say **without** running the AI, so the CPU barely warms up. Only use F6/F8 (polish) and F9/F10 (translate) when you actually need them — those are what run the AI model.
- **The heat is momentary.** The engine only works the CPU while it's processing (the few seconds after you stop a recording, or while it prepares speech). The rest of the time it sits idle and cool.
- **Still warm or slow?** Set WHISPER_MODEL_NAME="base" in .env (faster and cooler), then restart.
- **Keep .env light:** small + qwen2.5:3b. Avoid medium and qwen2.5:7b on a laptop — they're the usual cause of fan noise and heat.
- **Airflow:** use the laptop on a hard surface (not a bed or cushion) so the vents aren't blocked.

Making it faster or more accurate

After changing any of these in `.env`, **save the file and restart the engine** (close the console and run the launcher again).

Speech model (`WHISPER_MODEL_NAME`):

Value	Trade-off
<code>tiny / base</code>	Fastest, lowest accuracy. Good if "small" feels slow.
<code>small</code>	Recommended balance for a laptop (this guide's default).
<code>medium</code>	Better Arabic accuracy, noticeably slower on CPU.

AI model (`OLLAMA_MODEL_NAME` — download it first with `ollama pull <name>`):

Value	Trade-off
<code>gemma2:2b</code>	Lightest, fastest.
<code>qwen2.5:3b</code>	Recommended balance, good Arabic + English.
<code>qwen2.5:7b</code>	Best quality, slower on CPU.

PRIVACY Privacy: everything in this guide runs **locally on your machine** — your voice and text are never uploaded. For deeper customization (other languages, other AI providers, prompts), see **CONFIGURATION.md**.